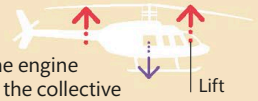
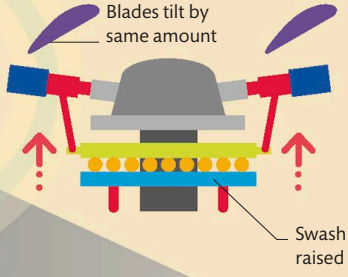
The collective and cyclic controls

To generate lift and change direction, the pilot uses the collective and cyclic controls. To increase or decrease lift, the collective lever raises or lowers the swash plate, changing the angle, or pitch, of all the blades at the same time. To change direction, the cyclic stick is used to tilt the swash plate, giving the blades unequal pitch depending on whether they are in front of or behind the rotor shaft.

KEY
- Lift
- Weight

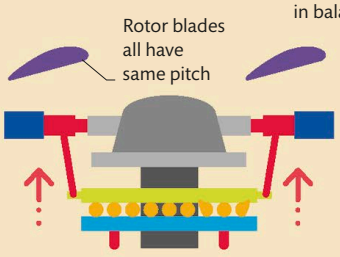
Takeoff
To take off, the pilot increases the engine speed and raises the collective lever to produce more lift.





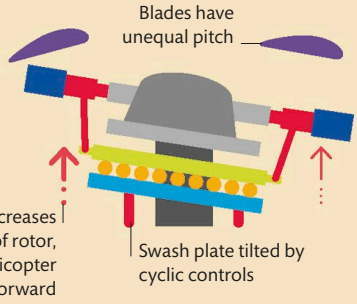
Hovering
To hover in one spot, the rotor blades produce just enough lift to balance the helicopter's weight.



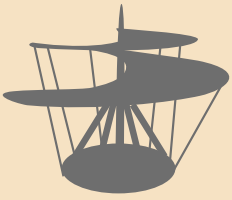


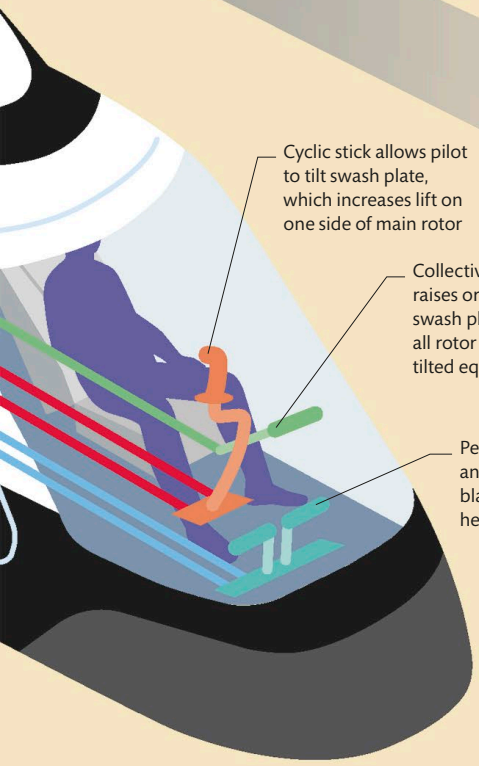
Forward flight
To go forward, the cyclic stick is pushed forward, which tilts the rotor up at the back.





IN 1480, LEONARDO DA VINCI SKETCHED AN IDEA FOR AN AIRCRAFT THAT COULD FLY VERTICALLY





TANDEM ROTOR BLADES

Instead of using a tail rotor to counteract torque, some helicopters have two overhead rotors that spin in opposite directions. The helicopter is steered by tilting the front rotor in one direction and the rear rotor in the opposite direction.

